

Math 122: Exam 1 - Practice 2

Student Name: _____

Instructions:

- Show all your work clearly. Full credit will only be given for complete and correct solutions with proper mathematical notation.
- Each question is worth 10 points.

Problem 1

Find the exact arc length of the curve over the interval from $x = 0$ to $x = 1$:

$$y = \frac{2}{3}(x^2 + 1)^{3/2}$$

Problem 2

Find the limit:

$$\lim_{x \rightarrow 0} \frac{\tan(2x)}{\ln(1+x)}$$

Problem 3

Find the limit:

$$\lim_{x \rightarrow +\infty} x^2 e^{-2x}$$

Problem 4

Find the limit:

$$\lim_{x \rightarrow 0^+} (1 + 2x)^{1/x}$$

Problem 5Find $\frac{dy}{dx}$:

$$y = \cos^{-1} \left(\frac{1}{x} \right)$$

Problem 6

Find $\frac{dy}{dx}$:

$$y = e^{2x} \tan^{-1}(x)$$

Problem 7

Evaluate the integral:

$$\int \frac{dx}{\sqrt{1-9x^2}}$$

Problem 8

Evaluate the integral:

$$\int \frac{\cos x \, dx}{\sqrt{1-\sin^2 x}}$$

Problem 9

Evaluate the definite integral:

$$\int_0^1 \frac{(\tan^{-1} x)^2}{1+x^2} dx$$

Problem 10

Evaluate the integral:

$$\int \frac{\cos \theta}{\sin^2 \theta + 1} d\theta$$

— END OF EXAMINATION —